

Creating Your Own Application in OpenDaylight

Hosted by the Linux Foundation, the OpenDaylight project was created to promote software defined networking (SDN) and to create a solid foundation for network functions virtualisation.

This article takes you through the steps needed to create an application in OpenDaylight.



OpenDaylight (ODL) is an open source software-defined networking (SDN) project hosted by the Linux Foundation. It was created in order to advance SDN adoption and build a strong base for network functions virtualisation (NFV). It is a modular platform, with most modules sharing some common services and interfaces. Each module is developed under a multi-vendor sub-project. You can find the list of projects at https://wiki.OpenDaylight.org/view/Project_list.

Technology stack

The technology stack consists of the following four layers.

1. **Maven:** OpenDaylight uses Maven for its complete build automation, and its archetype has been used for setting up the initial project structure to develop applications.
2. **Java:** The OpenDaylight framework is built in Java.
3. **Apache Karaf:** This is a lightweight OSGi container for loading modules/bundles dynamically at runtime.
4. **YANG:** OpenDaylight uses YANG for data modelling language, RPC and notifications between different modules.

MD-SAL: An overview

The Model Driven Service Abstraction Layer (MD-SAL) is the core of the OpenDaylight project. It helps in connecting different layers and modules through a well-defined API. The MD-SAL uses YANG as the modelling language for both interface and data definitions, and provides a messaging and data-centric runtime for such services based on YANG modelling. YANG tools are used to compile the YANG template and generate Java classes/interfaces, and automatically build a REST API Doc Explorer.

The flow of the application development process

The prerequisites are:

1. Maven 3.1.1 or later
2. Java 7 or later
3. Eclipse IDE and YANG plugin (optional)

Steps to develop an OpenDaylight controller application

Let's now list the nine steps that will help you build an OpenDaylight controller application for Beryllium release.